

Problem Statement ID- 26181

Problem Statement Title - A secure AI health companion for real-time monitoring, early risk detection, and disaster resilience through on-device intelligence.

Theme - MedTech / BioTech / HealthTech

PS Category - Hardware

Team Name- Braj Coders

Team ID - 1111

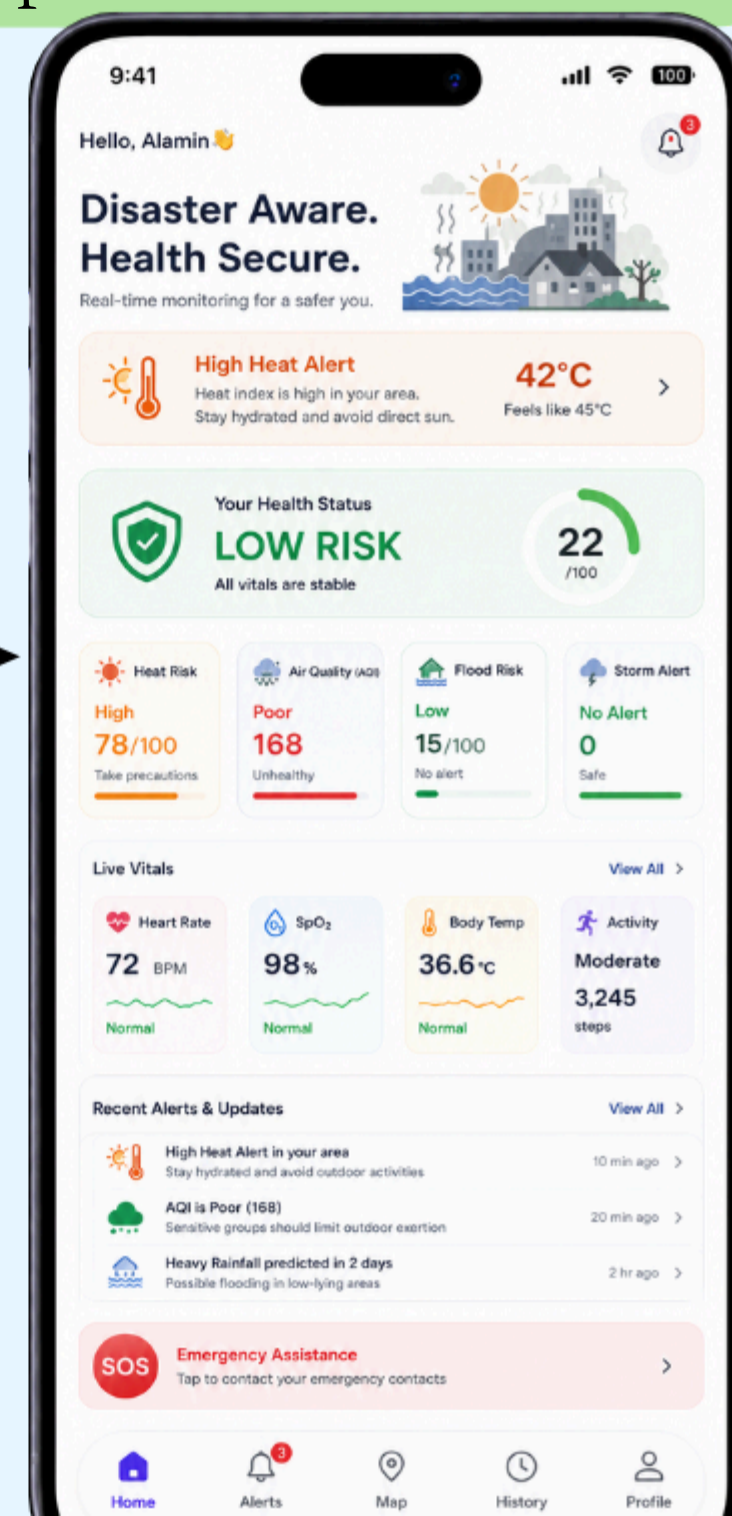
Team Members - Ritika Rawat, Sahil Chaudhary, Nikhil Singh, Pawan Kumar, Shubham Jain, Sweety



On-Device Intelligence Real-Time Protection.

“Develop a wearable that uses on-device AI to spot health risks early and stay useful during disasters.” SIH26181
Sanjeevni reads your heartbeat, your movement, and the air around you, and warns you before things turn serious. All on your phone. Nothing sent to the cloud. Built for India's heat waves, floods, and pollution.

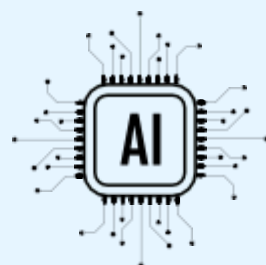
- ✓ Real heart signal, not guesswork: an actual ECG sensor, not just a pulse count.
- ✓ Every alert is decided on your phone, not a server somewhere.
- ✓ Knows the difference between "you're just exercising" and "something's wrong".
- ✓ Still works with no internet: sends an SOS by SMS if needed.
- ✓ Built into your clothing tracks your rest and sleep too, so you never have to remember to wear a separate device.



Real ECG hardware



Runs on your phone,
not the cloud



Reads context,
not just numbers



Works with
zero signal



Daily Wearable,
not a gadget



Personalizes over time

TIER 1 - SENSING LAYER

BioAmp (ECG/PPG/EMG)



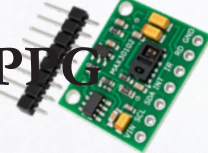
- Heart Rhythm
- HRV, Arrhythmia detection



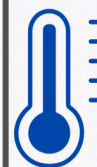
MAX30102,



- Heart rate
- SpO₂ + PPG
- Blood oxygen saturation



DHT22 / BME280



- Temperature + Humidity
- Heat index
- Dehydration risk



MQ135 / PM2.5 Sensor



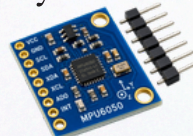
- Air Quality
- Pollution
- AQI index



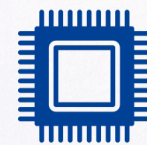
MPU06050



- Accelerometer + Gyro
- Fall detection
- Activity tracking



Environment Node



Sensor Integration Unit

ESP32



Sensor Hub

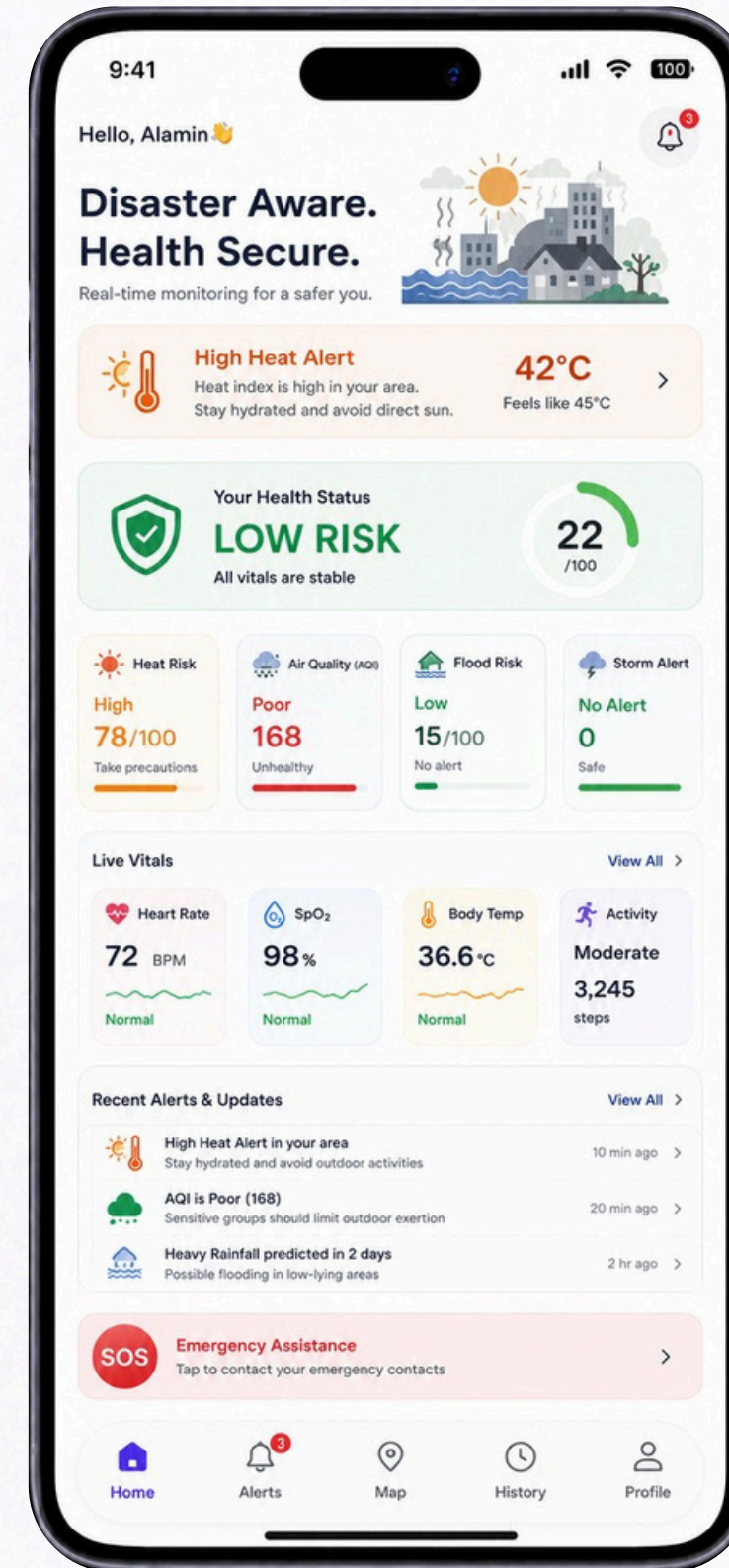
- Data acquisition
- Basic filtering
- Feature extraction
- BLE transmission



Motor / Actuator Control Unit

- Haptic Feedback
- Alert Vibration
- Buzzer / LED

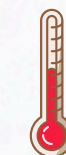
TIER 2- EDGE AI INFERENCE LAYER



HR:
72 BPM



SpO₂:
98%



Temp:
32 °C



AQI:
45

STATUS:
NORMAL

TIER 3- USER LAYER/ THE WEARABLE



ON DEVICE AI MODELS (TensorFlow Lite)

Fall Detection Model

Heart Health Model

Respiratory Risk Model

Heat Stress Model

ALERTS & ACTIONS



Location



Local Alert



SOS Call



Works without inter net

TECHNICAL FEASIBILITY:



- **Hardware & Connectivity:** Proven, BioAmp, MPU6050, PM2.5) and other sensors with ESP32 with stable BLE communication
- **On-Device AI:** TFLite runs locally via phone's NPU — no special hardware or internet needed

ECONOMIC FEASIBILITY:



- **Hardware Cost:** Prototype costs ₹1,500–2,500/unit — far cheaper than commercial wearables (₹5,000–20,000+)
- **Operating Cost:** No recurring cloud/API fees and open-source stack keep scaling costs near-zero

LEGAL FEASIBILITY:



- **Privacy & Compliance:** All raw health data stays on-device, aligning with data protection norms
- **Regulatory Scope:** No diagnostic claims at MVP stage positioned as risk- device avoiding medical regulations

Challenges

Garment movement and **loose contact cause noisy, unreliable ECG signals** during normal daily activity



Hard to distinguish genuine medical distress from harmless heart-rate spikes caused by physical exertion



Continuous monitoring across **ECG, motion, and air sensors drains battery fast**, limiting real-world usability



Rural and elderly users often **struggle with complex apps, setups, and unfamiliar interfaces**



Solutions

Motion-referenced **adaptive filtering** combined with **bandpass processing** cleans the ECG signal in real time

Multi-sensor fusion cross-checks heart-rate data against live motion context before triggering any alert

Duty-cycled sampling with **low-power BLE** and **optimized polling intervals** significantly **extends battery life**

Simple color-coded Green/Yellow/Red UI with regional-language **voice alerts** and **assisted setup**

The Sanjeevni



IMPACTS

Elderly & at-risk populations:

Continuous passive monitoring without needing to remember to wear/check a separate device.



Outdoor workers

Real-time heat-stress and air-quality risk alerts during exposure, potentially reducing heat-related occupational illness.



Rural/underserved communities

Functions fully offline brings health monitoring to areas with unreliable connectivity and limited healthcare access.

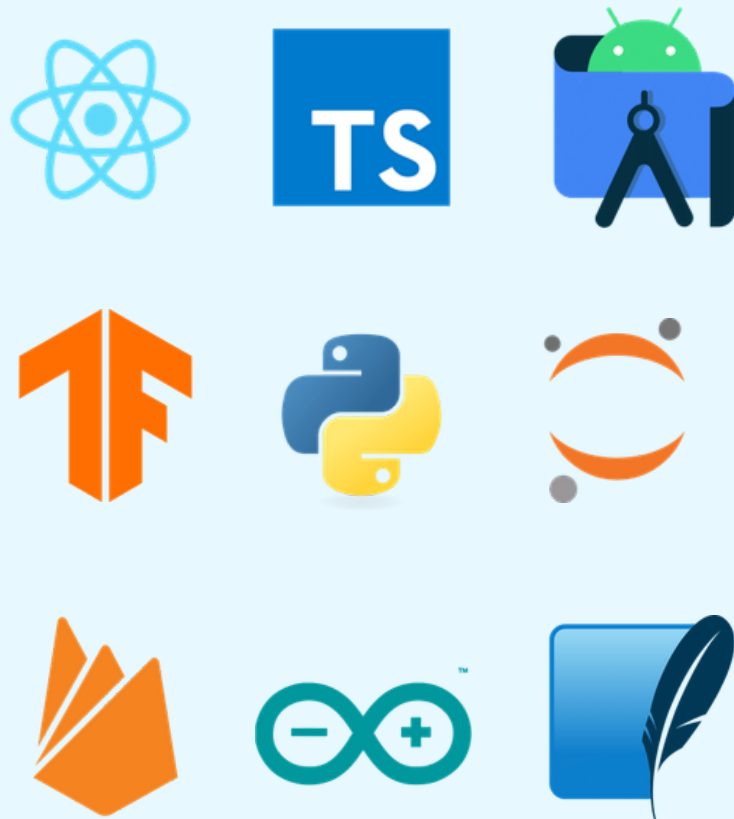


Disaster-response scenarios

Individual-level early warning during heat waves, floods, pollution events complements, doesn't replace, public health infrastructure.



TECH STACK



BENEFITS

Privacy-first

Raw health data never leaves the device a meaningfully different trust model than cloudbased health apps.



Low-cost & scalable

Sub₹ 2,500 hardware cost per unit enables deployment at population scale, not just premium-consumer scale.



Works without infrastructure

No internet or cellular data required for core monitoring and SOS resilient exactly when infrastructure is most likely to fail (disasters).



Early warning, not just tracking

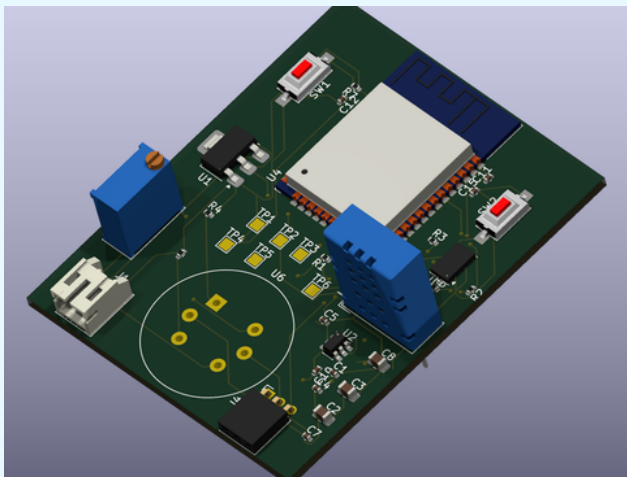
Moves from passive data logging to active risk alerting and actionable recommendations



References

- MIT-BIH Arrhythmia Database (PhysioNet) — for ECG anomaly model training reference
- Public HAR datasets (WISDM/UCI HAR/PAMAP2) — for activity classification reference
- IMD heat-wave criteria/advisories — for regional heat-index thresholds CPCB
- National AQI standards — for air-quality risk banding

Own Custom PCB Design



Evaluating key features

Feature	Sanjeevni	Generic Fitness Band (Mi Band/Fire-Boltt class)	Apple Watch/Premium Wearable
Real ECG (not just PPG)	✓	✗	✓ (limited, requires activegesture)
On-device AI (no cloud)	✓	✗	Partial
Works fully offline (SOS)	✓	✗	✗ (needs paired phone + data)
Air quality sensing	✓	✗	✗
Disaster-specific alerts (heat/AQI)	✓	✗	✗
Cost	Low (~₹1,500-2,500 prototype)	Low-Medium	Very High